# Smart traffic lights that learn

A systematic literature review of reinforcement learning for traffic signal control: from simulation to real streets

Rakhimzhanov Amangeldy · Nurkin Aidar · Midterm progress report

# Why traffic lights should learn

## Fixed-time light

Same green and red durations whether the road is empty or jammed

## Learning light

Watches real traffic and changes its timing on the fly

Hundreds of studies report shorter waits **in simulation**, yet peer-reviewed tests on **real streets** are still very rare. Our review asks why this gap exists.

# A smart system in a loop

01

## Sense

Cameras and road sensors measure queues and waiting cars

02

## Decide

An RL agent picks the next green phase

03

## Act

The signal switches on the real road

04

## Learn

Less waiting earns a reward, so the policy improves

> The light learns its timing by **trial and error**, not from a hand-made plan. Mistakes during learning are why real streets are hard.

# Six research questions

| Question | What we ask | What we extract |
|---|---|---|
| RQ1 Landscape | How fast is the field growing, and where? | Year, venue, country |
| RQ2 Methods | Which RL methods, states, actions and rewards? | Algorithm, agents, reward |
| RQ3 Validation | Tested only in simulators, or on real roads? | Simulator, data, realism level |
| RQ4 Performance | How much do waits, queues and travel times drop? | Values against baselines |
| RQ5 Barriers | What stops RL lights from reaching real cities? | Barriers and proposed fixes |
| RQ6 Future | Which gaps remain, and what should come next? | Gap matrix |

# How we review: PRISMA 2020

The international standard for systematic reviews, in six stages.

01

## Plan

Protocol, questions and criteria fixed before searching

02

## Search

Scopus, Web of Science and IEEE Xplore, 2019–2026

03

## Screen

Two reviewers: title and abstract, then full text

04

## Assess quality

Five criteria scored 0–2, maximum 10 points

05

## Extract

One codebook; every number traced to its table or page

06

## Synthesise

Taxonomy, statistics, gaps and the PRISMA flow diagram

# Search strategy

CONCEPT A · METHOD

## Reinforcement learning

Q-learning · deep Q-network · actor-critic · policy gradient · PPO

AND

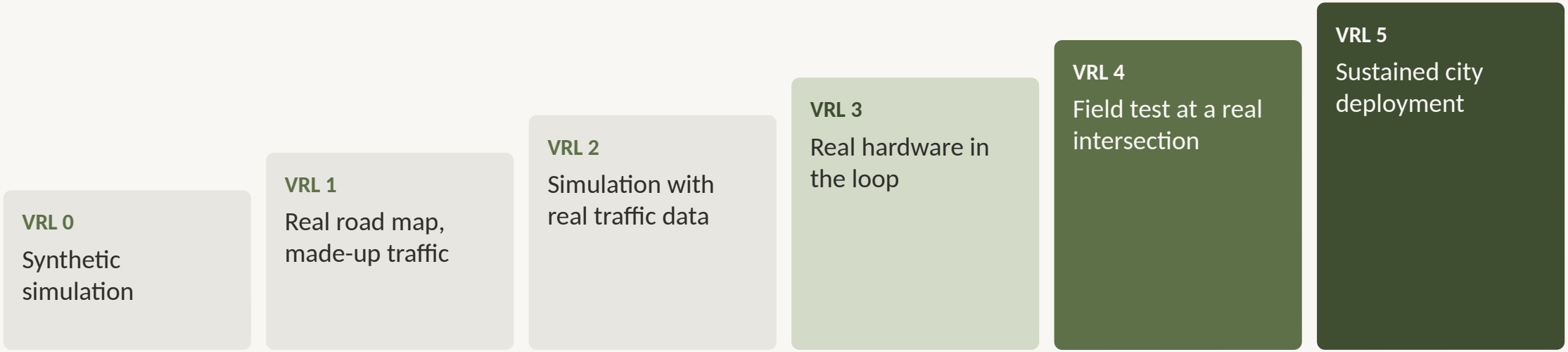
CONCEPT B · APPLICATION

## Traffic signals

traffic light · signalized intersection · signal timing · signal phase

| 3 | 2019–2026 | 20 |
| --- | --- | --- |
| databases searched | publication window | known papers the pilot must find |

# Our key idea: a validation realism scale




Every study gets the highest level it reaches. Then we ask: **do the reported gains shrink as tests get more real?**

# What the scoping search already shows

## Simulators dominate

SUMO and CityFlow are the main test platforms in recent reviews

## Field tests are rare

Deployment-focused papers from 2021–2022 report almost no live RL signals

## The last SLR stops in 2020

Noaeen et al. (2022) reviewed 160 papers up to March 2020

## Same data, many papers

Jinan, Hangzhou and New York datasets are reused again and again

Exploratory scoping search only: it shaped our design and is not counted as review evidence.

# What we have built so far

**15**

planning and method documents, one 32-page dossier

**3**

database search strings, ready to run

**64**

extraction fields in one codebook

**10**

pipeline commands, from import to PRISMA diagram

> Every PRISMA count and statistic will be **computed from the records**, never typed by hand. All 10 automated tests pass.

# Progress: about 50% complete

50%

The design half is done. The execution half starts with the database searches.

| Phase | Weight | Status | Earned |
|---|---|---|---|
| Planning and protocol (Steps 1–6) | 20% | Done | 20 |
| Search strategy (Steps 7–12) | 15% | Pilot and lock left | 13 |
| Review tools (Steps 13–27) | 15% | Built, tests pass | 15 |
| Database search and screening | 20% | Next | 0 |
| Quality and data extraction | 15% | Not started | 0 |
| Synthesis and writing | 15% | Taxonomy drafted | 2 |
| **Total** | **100%** | | **50** |